MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
Model Question Paper
MG4DSECHE201 – FOOD CHEMISTRY
(2024 ADMISSION ONWARDS)

Duration: 2 hrs Max. Marks: 70

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions. Each question carries **2** marks

1. Name two water soluble and two fat soluble vitamins [K [CO: 3]
2. What are the health benefits of pepper? [U] [CO:4]
3. What are the different states of water present in foods? [K] [CO: 2]
4. What is fermentation? [U] [CO: 4]
5. Distinguish between modified and controlled atmosphere packaging (MAP & CAP) [U] [CO:1]
6. What are macrominerals? Give two examples. [K] [CO: 3]
7. What are leavening agents? Give two examples [K] [CO: 1]
8. Give two symptoms and two causes for vitamin toxicity in the body [K] [CO: 3]
9. What do you meant by lipid oxidation? [U] [CO: 2]
10. What are the components present in cardamom? [U] [CO:4]
11. What do you understand by the term pasteurization? Mention its objectives [U] CO: 1]
12. Why is fructose sweeter than glucose? [U] [CO: 2]

(10∗2=20)

Part B

Short Essay Type questions

Answer any **6** questions. Each question carries **5** marks

13. Describe the causes of variation and losses of vitamins in foods. Explain how processing, storage and cooking affect vitamin stability [U][CO:3]

14. How the chemical components of Ginger contributes to its medicinal and culinary uses. [A][CO:4]

15. Describe the properties and applications of sugar derivatives in food processing. [U] [CO: 2]

16. Analyse the consequences of deficiency of the following dietary minerals on human metabolism and physiological functions. a) Iron b) Zinc c)Iodine d) Selenium and e) Copper [An][CO:3]

17. Write the main functions of the Food Safety and Standards Authority of India (FSSAI) [U] [CO:1]

18.List out four health benefits of moderate consumption of wine. [U] [CO:4]

19. What are artificial sweeteners? Discuss the properties and chemical structure of any two artificial sweeteners [U] [CO:1]

20. Explain the principle of water activity and its importance in foods? [U] [CO: 2]

(6*5=30)

Part C

Essay Type questions

Answer any **2** questions. Each question carries **10** marks

21. Discuss the role of proteins in human nutrition and food product development. [U] [CO: 2]

22. Describe the societal role of food chemists in ensuring food safety, quality, nutrition, adulteration control, and public health. [U][CO:3]

23. Write a note on the Adulteration of herbs and spices. Also describe various methods for the determination of adulterant. [U] [CO:4]

24. What are food additives? Discuss any 4 types of food additives and their functions with examples. [U] [CO:1]

(2*10=20)